Muhammad Rafi Arsya

Software Engineering Student

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Profile

Software Engineering student at Universiti Malaya who takes ideas all the way to live products. Built and runs two student platforms, CampusBay and UMOVE, end to end: Next.js and React front ends, Node and PostgreSQL back ends, Cloudflare Workers, and a self-hosted Linux server behind a Cloudflare Tunnel. Also works in applied AI/ML (RAG, CNNs written from scratch). Seeking a Data Engineering or Software Engineering internship.

Education

Universiti Malaya (UM), Malaysia *Since 2025*
B.Sc. Computer Science (Software Engineering)

- Coursework: Data Structures & Algorithms, Machine Learning, Network Technology, Human Computer Interaction, Computer Systems Architecture.

International Program ICBS, Riau, Indonesia *2022 to 2025*
Senior High School

- International curriculum with English-medium instruction.
- Active in student leadership and language programs.

Experience

PEKOM: Computer Science Society, Universiti Malaya *Since 2026*
Technical Executive Committee

- Serving in the Website Division, building and maintaining the society's web platforms.
- Supporting faculty-wide events across the Faculty of Computer Science & IT.

PPI Malaysia (Indonesian Students Association) *Since 2025*
Head of Website Division, Data & Information System Bureau

- Full ownership of the official PPI Malaysia website: development, maintenance, and continuous improvement.
- Serving Indonesian students across Malaysia.

PPI University of Malaya *Since 2026*
Head of Department, Art Exhibition at IDFEST

- Directing the Art Exhibition division for IDFEST, an international event showcasing Indonesian heritage at Universiti Malaya, from concept and curation through to delivery.
- Leading the department end to end: theme and curation, artist and submission coordination, exhibition layout, and delivery on event day.

PPI University of Malaya *Since Jun 2026*
Website Development Committee at IDFEST

- Building the official IDFEST website with Next.js 16, React 19 and Tailwind CSS v4, translating Figma designs into production-ready pages.
- Owning the homepage sections and animated timeline, hero responsiveness across breakpoints, and shared navigation and mascot components reused by the Art Exhibition and Musical Theater pages.

Leadership & Activities

Institut Teknologi Batam (ITEBA), Indonesia *October 2026*
Student Representative & Group Leader, Outbound Programme

- Represented Universiti Malaya as its student representative: led a group of 28 students and delivered the student speech on stage at ITEBA, Batam, during a programme that included the public lecture "Behind the Scenes of TikTok & ChatGPT".

LARAS, PPI Malaysia *2026*
Transportation Team & Field Committee

- Handled logistics, on-ground coordination and event-day operations for LARAS, PPI Malaysia's annual gathering.

Volunteering

Universiti Malaya Week of Welcome (UM-WOW) *2026*
Buddy, Student Affairs Department

- Welcomed and guided new students through orientation week, registration and their first days on campus at Universiti Malaya.

Faculty of Computer Science & Information Technology, Universiti Malaya *1 to 5 September 2026*
Student Buddy, Inbound Mobility Programme 2026

- One of a four-person committee running a five-day FSKTM programme hosting students from Chulalongkorn University, Thailand's top-ranked university, in Kuala Lumpur.
- Served as a Student Buddy to the visiting delegation and was responsible for documentation across the whole programme, covering sessions and daily activities on all five days.

Selected Projects

CampusBay: Student Marketplace *2026*
Live marketplace where students list items or post requests and close deals on WhatsApp. Next.js 16 on Cloudflare Workers with D1 and Drizzle, instant updates through a Durable Object, EN/MS/ID search and a full moderation panel. Built and run solo.

UMOVE: Campus Errand Delivery *2026*
Students post errands and verified student runners deliver them, paid on delivery. React 19 front end on Cloudflare Workers; Node, Hono and PostgreSQL API in Docker on a mini PC behind Cloudflare Tunnel, with a live board over Server-Sent Events.

PaperMind: Local RAG Paper Analyzer *2026*
Ask questions of any PDF with answers grounded in the text. FastAPI, ChromaDB and Ollama, fully offline, in Docker Compose on a self-hosted Linux server.

AI Digit Recognizer: CNN From Scratch *2026*
Convolutional network written entirely in NumPy, from convolutions to backpropagation and Adam. 99.1% on the MNIST test set, served by Flask with a layer-by-layer visualizer.

Nalar: Blog with a Custom CMS *2026*
Node.js, Express and PostgreSQL blog with JWT auth, Markdown editing and scheduled publishing. Self-hosted behind Cloudflare Tunnel, managed with PM2.

snip: Serverless URL Shortener *2026*
Click-tracking URL shortener on API Gateway, Lambda and DynamoDB, defined as infrastructure as code with AWS SAM and least-privilege IAM per function.

Crop Disease Detector *2026*
Classifies 38 plant diseases from a leaf photo. MobileNetV2 transfer learning with two-phase fine-tuning in TensorFlow/Keras, about 97% accuracy, deployed on Hugging Face Spaces.

IDFEST 2026 Website *2026*
Official site for PPI UM's cultural festival, built as a team in Next.js 16 and Tailwind v4. Owned the homepage timeline, responsive hero and shared navigation.

Skills

Core: TypeScript, JavaScript, Python, React, Next.js, Node.js, PostgreSQL / SQL, Docker, Cloudflare, Git

Also used in projects:

Languages: Java, PHP, Bash

Frontend: Tailwind CSS, Vite, Bootstrap 5, Chart.js / Recharts, PWA, Chrome Extensions (Manifest V3)

Backend: Express, Hono, FastAPI, Flask, WebSockets / Socket.IO, Server-Sent Events, Cloudflare Workers & Durable Objects, Zod, JWT & Google OAuth

Data: SQLite / Cloudflare D1, Drizzle ORM, Supabase, DynamoDB, ChromaDB

Infrastructure & Testing: Docker Compose, Cloudflare Tunnel, AWS (Lambda, API Gateway, SAM), NGINX, Ubuntu self-hosting, GitHub Actions CI, Playwright

AI / ML: NumPy, TensorFlow / Keras, OpenCV, MediaPipe, scikit-learn, pandas, RAG with Ollama, Hugging Face, Claude API

Certifications & Courses

- Ultimate AWS Certified Developer Associate (DVA-C02), Udemy course completion, Stephane Maarek, June 2026
- Ultimate AWS Certified AI Practitioner (AIF-C01), Udemy course completion, Stephane Maarek, May 2026
- Foundations: Data, Data, Everywhere, Google Data Analytics Certificate, Coursera, September 2026
- IELTS Academic, Band 5.5 (CEFR B2), British Council IDP, June 2025
- Additional coursework, around 570 hours across 2025 and 2026: freeCodeCamp JavaScript Algorithms & Data Structures (300 hours), The Complete JavaScript Course (Jonas Schmedtmann), full-stack bootcamps with Angela Yu and Cloudaffle, Advanced SQL (Maven Analytics), The Complete SQL Bootcamp (Jose Portilla), Spring AI Fast Track, and Cloud & Gen AI on AWS plus Spec-Driven Development with Kiro (Dicoding)

Languages

Bahasa Indonesia: Native | English: Professional working proficiency | Malay: Professional working proficiency